

Assignment :- 4

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① Define unstructured data with examples.

→ unstructured data doesn't follow a fixed format or schema.

Examples include emails, videos, images, text files, and social media posts.

② What are the main challenges in managing unstructured data?

→ Difficult to store, search, and analyze efficiently.

Requires complex tools like NLP and AI for processing.

③ Explain the Boolean retrieval model.

→ uses logical operators like AND, OR and NOT to match documents with queries.

Returns exact matches based on the query's Boolean condition.

④ What is the purpose of tokenization in text processing?

→ It splits text into smaller units called tokens (words or phrases).

This helps in text analysis, indexing, and NLP tasks.

⑤ Differentiate between stemming and lemmatization.

- Stemming removes suffixes to get root forms (e.g. "Planning" → "play")
- Lemmatization uses linguistic rules to find valid base words (e.g. better → good)

⑥ Describe the vector space model

- Represents documents and queries as vectors in multidimensional space.
- Relevance is measured using similarity measures like cosine similarity

⑦ What is the significance of stop word removal?

- Removes common words like "the", "is", and "of" to reduce noise.
- Improves efficiency and focus on meaningful terms in text analysis.

⑧ Explain the TF-IDF weighting scheme.

- Term Frequency - Inverse Document Frequency measures term importance.
- High TF-IDF means a term is frequent in one document but rare across others.

⑨ What is an inverted index and how does it work?

→ It maps each term to a list of documents containing it. used in search engines to quickly retrieve relevant results.

⑩ Describe the PageRank algorithm.

→ Ranks web pages based on the no and quality of links to them. Pages linked by many high-ranking pages get higher importance.

⑪ What are the advantages of word embeddings?

→ They capture semantic meaning and relationships between words. Enable better similarity and context understanding in NLP models.

⑫ Explain precision and recall in IR evaluation.

→ Precision = relevant retrieved / total retrieved.

Recall = relevant retrieved / total relevant.

Both measure retrieval accuracy.

⑬ What is query expansion and why is it used?

→ Adds related terms to a query to improve recall.

Helps capture synonyms or related concepts the user didn't include.

⑭ Describe the probabilistic retrieval model.

- Assigns a probability that a document is relevant to a query.
- Retrieves documents with the highest estimated relevance score.

⑮ What is Latent Semantic Indexing (LSI)?

- Uses Singular Value Decomposition to detect hidden patterns in term-document data. Groups semantically related terms and documents together.

⑯ Explain document clustering techniques.

- Groups similar documents into clusters without labels.
- Common methods include K-means and hierarchical clustering.

⑰ What are n-grams and their applications?

- Sequences of n consecutive words or characters (e.g. "data science") used in text prediction, language modeling, and plagiarism detection.

⑱ Describe the BM25 ranking function.

- An improved TF-IDF model that adjusts for document length and term saturation.
- used in modern search engines for ranking results.

⑮ Describe the role of cosine similarity in document retrieval?

→ measures the angle between document and query vectors.
Higher cosine value means higher similarity and relevance.

⑯ Explain the concept of text normalization.

→ converts text into a consistent form. Lowercasing, removing punctuation, etc. Improves matching accuracy during text comparison.

⑰ What are the components of a typical IR system?

→ Document collection, indexing, query processing, ranking, and evaluation. Together, they manage, search, and retrieve relevant information.

⑱ Differentiate between supervised and unsupervised text classification?

→ Supervised uses labeled data for training; unsupervised doesn't.

Unsupervised groups data by similarity, not predefined categories.

(23) Describe the Rocchio algorithm.

→ Refines user queries using relevance feedback. Adjusts query vectors toward relevant documents and away from irrelevant ones.

(24) What is topic modeling in text analysis?

→ Discovers hidden themes within a large document set. Common methods include Latent Dirichlet allocation (LDA).

(25) Explain the concepts of document frequency.

→ It counts how many documents contain a specific term. Helps in weighting terms for relevance scoring like TF-IDF.

(26) What are word embeddings and their advantages?

→ Dense vector representations capturing meanings and context of words. Improve performance in NLP tasks like similarity and translation.

(27) Describe the language modeling approach to IR.

→ Builds a probabilistic model of each document to predict query likelihood. Documents are ranked by the probability of generating the query.

②⑧ What is the purpose of singular value decompositions in LSI?

→ Reduces the dimensionality of term-document matrices. Reveals underlying semantic relationship between words and documents.

②⑨ Explain the concept of term frequency.

→ Number of times a term appears in a document. used to measure term importance in models like TF-IDF.

③⑥ What are the limitations of Boolean retrieval?

→ Requires exact matches and offers no ranking. users must craft complex queries manually for accurate results.

③① Describe the k-means algorithm for document clustering.

→ Partitions documents into k clusters based on feature similarity. minimizes distance between points and their cluster centers.

③② What is sentiment analysis in text processing?

→ Determine the emotional tone of text: positive, negative, or neutral. used in reviews, social media, and opinion mining.

③② Explain the concept of document ranking.

→ Orders retrieved documents by relevance to the query. uses scoring functions like TF-IDF, BM25, or PageRank.

③④ What are the applications of named entity recognition (NER)?

→ Identifies entities like names, dates and locations in text.

used in information extraction, QA systems, and text summarization.

③⑤ Describe the process of text classification?

→ Assign documents to predefined categories based on content. uses models like Naive Bayes, SVM, or neural networks.

③⑥ What is the difference between information retrieval and data retrieval?

→ Data retrieval finds exact matches from structured data. Information retrieval deals with unstructured data and ranks by relevance.

③⑦ Explain the concept of relevance feedback.

→ System refines future search results based on user judgements of retrieved documents. Improve precision by learning user intent.

③ what are the challenges in multilingual information retrieval?

→ Different languages, scripts and semantics make matching complex. Requires translation, cross-lingual indexing, and language detection.

④ Describe the process of building an inverted index.

→ Extract terms, normalize them, and map each to documents containing them - stores term-document pairs for fast search retrieval.

⑤ what is the role of machine learning in text analysis?

→ ML automates feature extraction, classification, and sentiment detection. Improves accuracy and scalability of NLP and IR tasks.

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